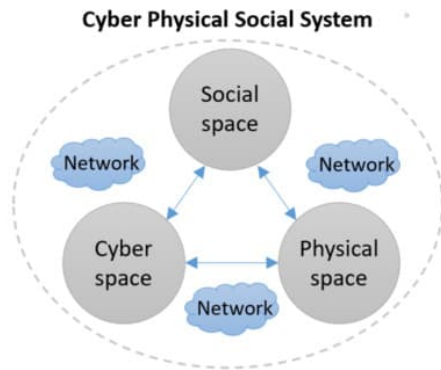
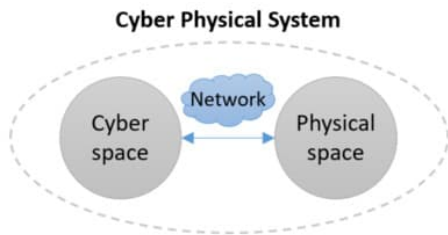


# Algorithmic Game Theory

University of Karaj

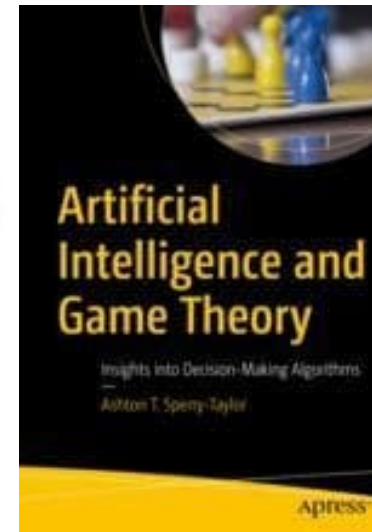
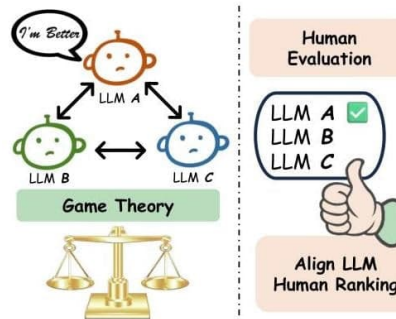
Dr. Amir Noori

Fall-2026



# Internet Economics

## Understanding Digital and New Media Markets



# Course Info

## References:

- [Twenty Lectures in Algorithmic Game Theory](#), Cambridge University Press, 2016.
- [Lectures in game theory for computer scientists](#) Cambridge University Press, 2011.

## Evaluation:

- Exam 60%
- Exercise 20%
- Seminar 20%

Email: [acnoori \(@\) gmail.com](mailto:acnoori@gmail.com)

Web: [acnoori.github.io](https://acnoori.github.io)

# Course Materials

- *Twenty Lectures on Algorithmic Game Theory.*
  - T. Roughgarden.
  - Cambridge University Press, 2016.
- *Lectures in game theory for computer scientists.*
  - K. R. Apt and E. Grädel.
  - Cambridge University Press, 2011.
- *Multiagent systems: Algorithmic, game-theoretic, and logical foundations.*
  - Y. Shoham and K. Leyton-Brown.
  - Cambridge University Press, 2008.
- *Algorithmic game theory.*
  - N. Nisan, T. Roughgarden, E. Tardos, and V. V. Vazirani.
  - Cambridge University Press, 2007.

# Algorithmic Game Theory

- **Algorithmic game theory** is an area that joins computer science and game theory to study **how smart, selfish groups** make **choices** and **share resources**.



Noam Nisan



Tim Roughgarden,  
(Stanford CS364A)



Erich Grädel  
(RWTH, Germany)



Kevin Leyton-  
Brown (UBC)



MohammadTaghi Hajiaghayi  
(Maryland)



Vahab Mirrokni  
(Google)



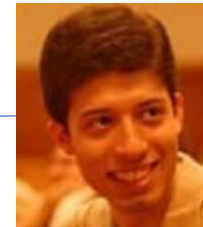
Eva Tardos,  
(Cornell)



Vijay Vazirani  
(U of California, Irvine)



Yoav Shoham  
(Stanford)

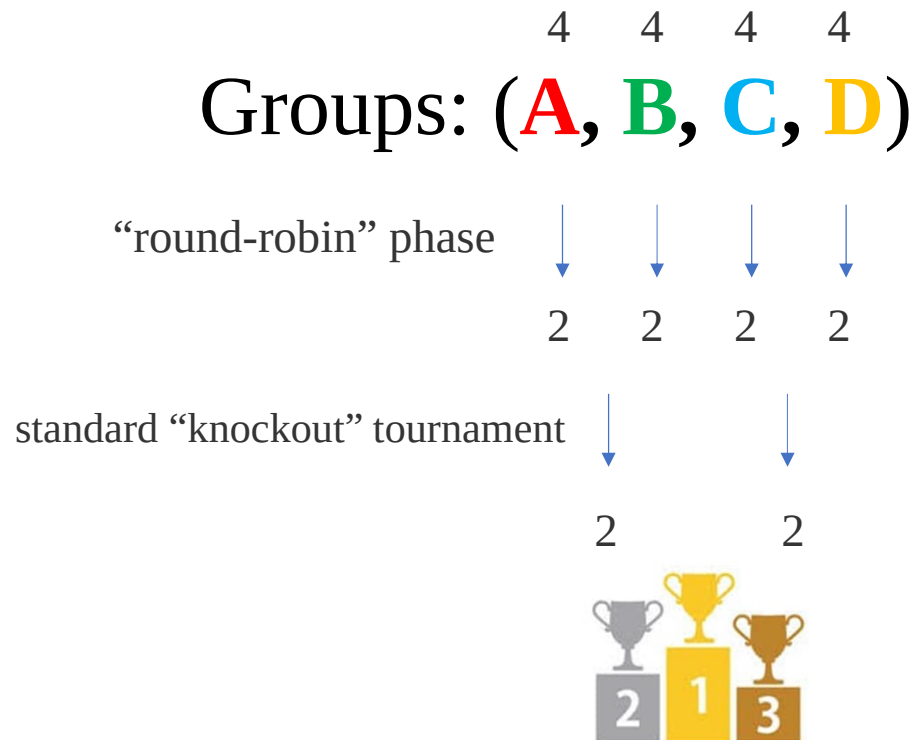


Mohammad Hossein Bateni  
(Google)

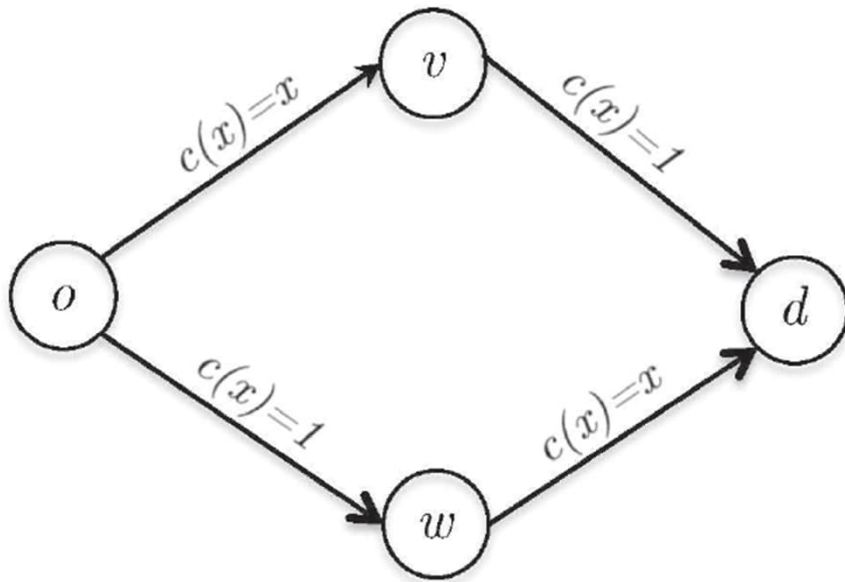
- Market design
- Mechanism Design
- Network Design
- Social Networks
- Scalable Algorithms
- Electronic Commerce
- Internet Economics
- Auctions

How Graphs Can Help You Stay Informed in an Evolving World, ACM26

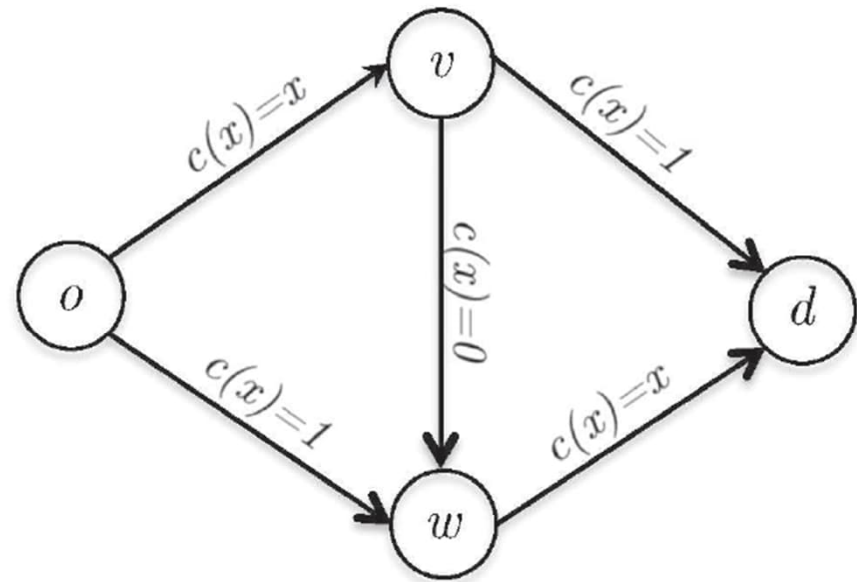
# Example (London Olympic 2012: a failed tournament design)



# Example (Networks)

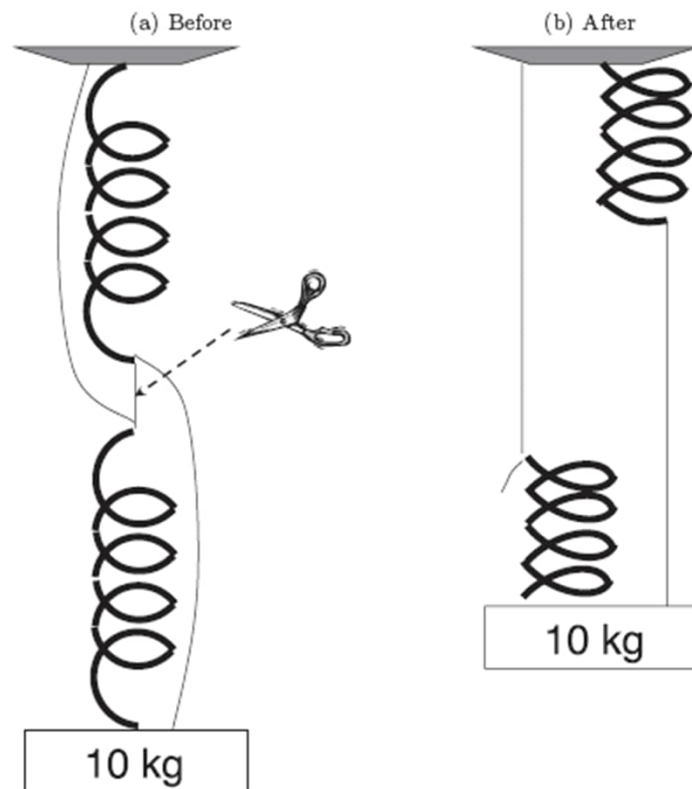


(a) Initial network



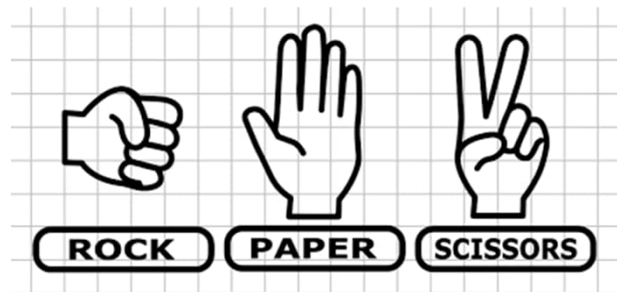
(b) Augmented network

# Example (Physics)



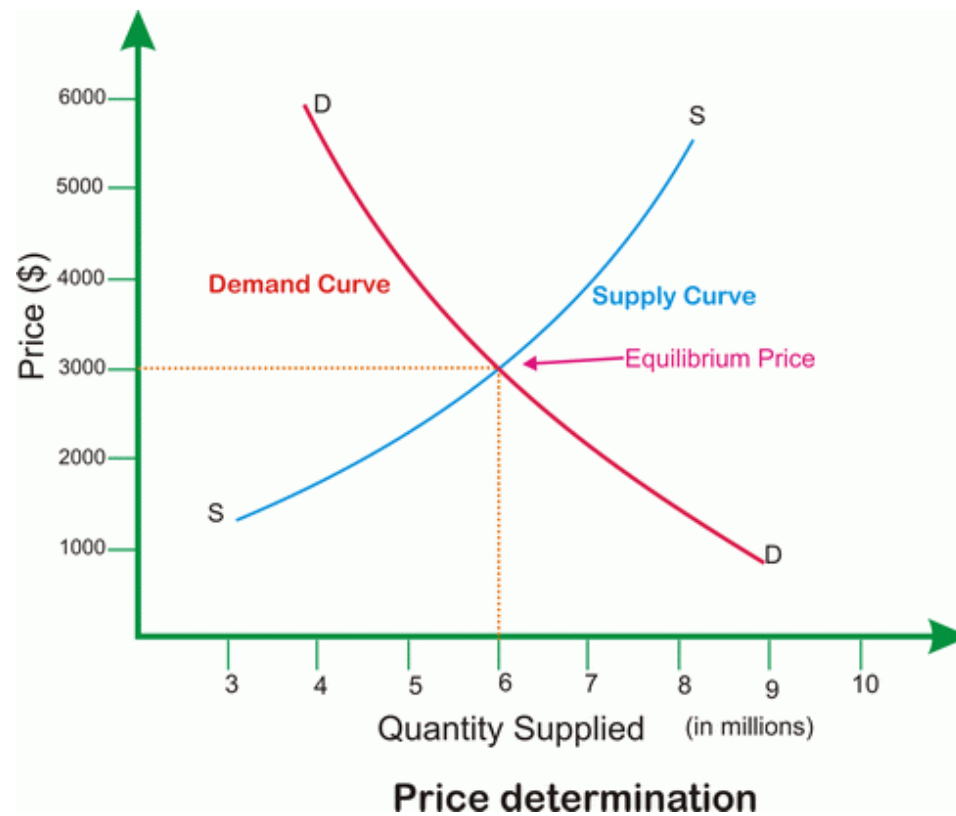
# Example (Zero-sum Games)

## Zero-Sum Games

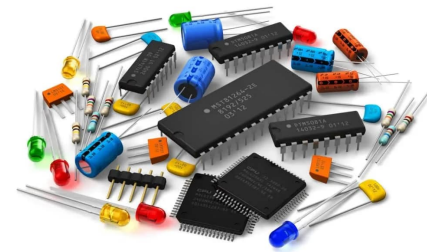


|          | Rock  | Paper | Scissors |
|----------|-------|-------|----------|
| Rock     | 0, 0  | -1, 1 | 1, -1    |
| Paper    | 1, -1 | 0, 0  | -1, 1    |
| Scissors | -1, 1 | 1, -1 | 0, 0     |

# Example (Supply and Demand)



e.g. Electronics



# Strategic Rule-Making (The Consequences of Selfish Behavior)

- **Mechanism Design**: Intersection of game theory, economics, and computer science to explain how rules influence human behavior.
- **Braess's paradox**, demonstrating how adding resources to a network can unintentionally worsen performance when participants act in their own interest.
- **Price of Anarchy** as the gap between selfish outcomes and optimal system performance.
- **Nash equilibrium** and the significant computational challenges involved in predicting stable states in complex games.

# Core Concepts

- **Games**: Situations where many independent players make choices that affect everyone else.
- **Equilibrium**: A stable state where no player wants to change their choice because they cannot do any better.
- **Mechanism Design**: Creating rules or algorithms so people are honest even when they want to win.
- **Price of Anarchy**: A number that shows how much worse a system works when everyone is selfish instead of helpful.

# Main Uses

- **Auctions**: Setting up online sales like Google ads or radio frequencies so people bid real prices.
- **Routing**: Helping data move across the internet without getting stuck in jammed paths.
- **Resource Sharing**: Splitting fair goods or tasks among many users.